Stereo II

Prof. Ziyun (Claude) Wang
Department of Electrical and Computer Engineering
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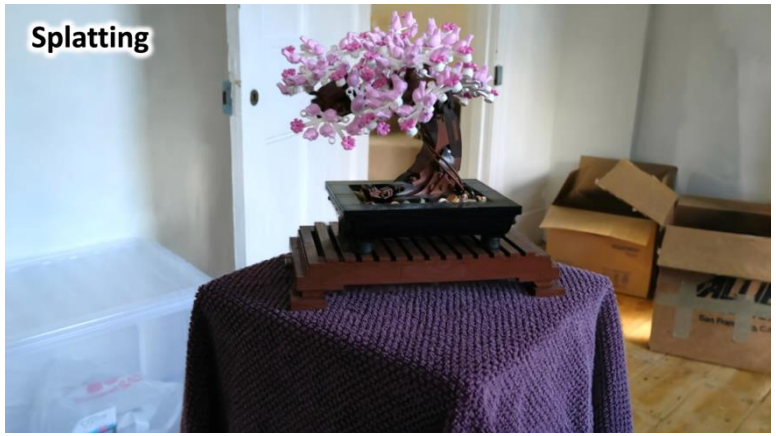
Logistics

- HW2 was released yesterday, due in two weeks (03/09)
  - Final optimization can take 1 to 2 hours

Recap: Gaussian Splatting

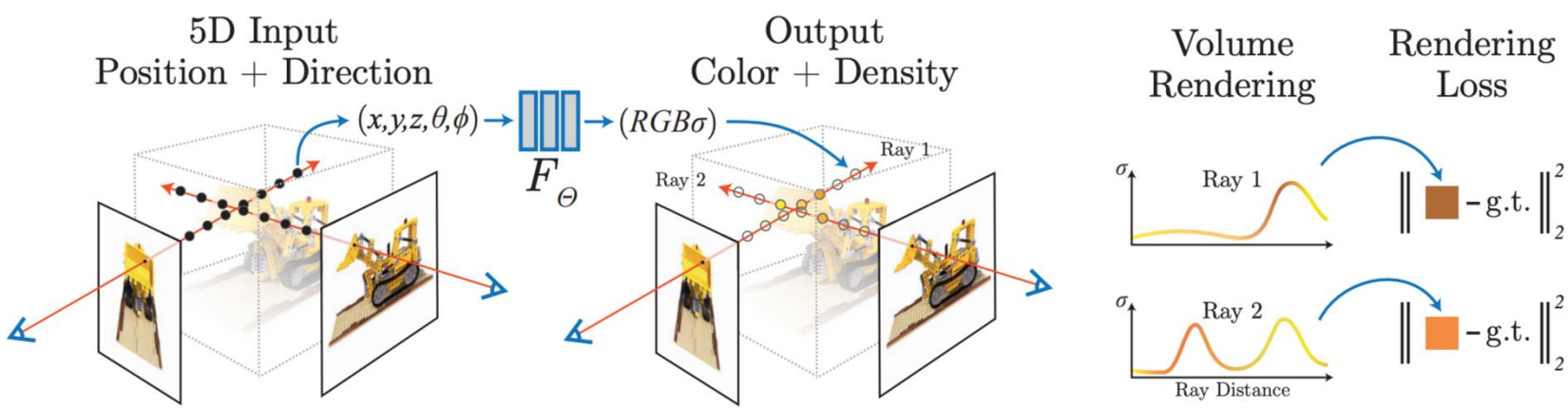


Novel View Synthesis (HW2)

Neural Radiance Field (NeRF)

A **neural radiance field** represents a scene as a continuous function implemented by an MLP:

$$f_\theta : (\boldsymbol{x} \in \mathbb{R}^3,\ \boldsymbol{d} \in \mathbb{R}^3) \mapsto \big(\sigma(\boldsymbol{x}),\ \boldsymbol{c}(\boldsymbol{x}, \boldsymbol{d})\big).$$



Modeling Continuous Radiance Field

For a camera ray parameterized as

$$\boldsymbol{x}(t) = \boldsymbol{o} + t\,\boldsymbol{d}, \qquad t \in [t_n, t_f],$$

$$\boldsymbol{C}(\boldsymbol{o}, \boldsymbol{d}) = \int_{t_n}^{t_f} T(t)\,\sigma(\boldsymbol{x}(t))\,\boldsymbol{c}(\boldsymbol{x}(t), \boldsymbol{d})\,dt$$

$$\boxed{T(t) = \exp\left(-\int_{t_n}^{t} \sigma(\boldsymbol{x}(s))\,ds\right)}$$

- $T(t)$: visibility / accumulated "not yet occluded" probability
- $\sigma(\boldsymbol{x})$: density (controls how quickly visibility decays)
- $\boldsymbol{c}(\boldsymbol{x}, \boldsymbol{d})$: radiance/color contributed at location $\boldsymbol{x}$ toward view direction $\boldsymbol{d}$

Discretize along the ray

In practice, NeRF uses a sampled approximation with points $\{t_k\}_{k=1}^{K}$ and step sizes $\delta_k = t_{k+1} - t_k$. Let

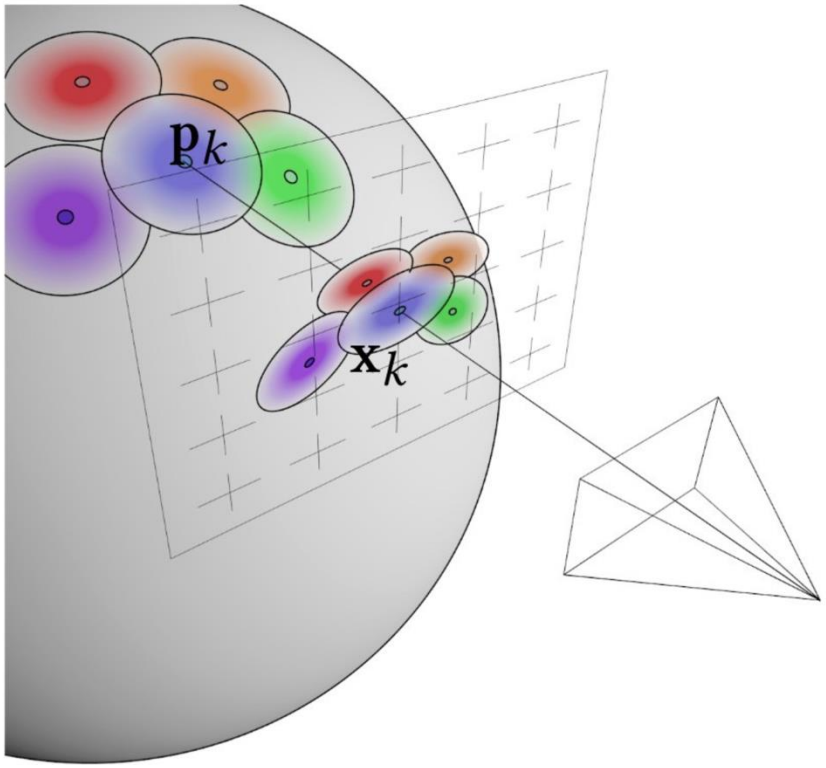
$$\sigma_k = \sigma(\boldsymbol{x}(t_k)), \qquad \boldsymbol{c}_k = \boldsymbol{c}(\boldsymbol{x}(t_k), \boldsymbol{d}).$$

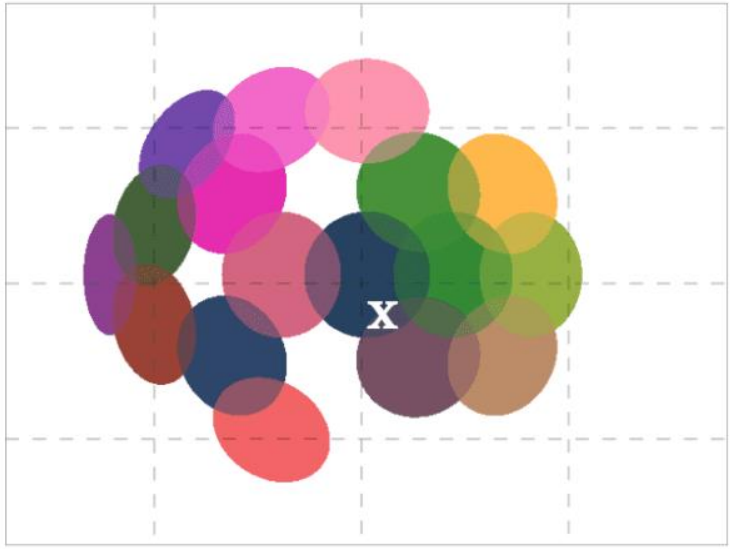
Define per-sample opacity and weights:

$$\alpha_k = 1 - \exp(-\sigma_k \, \delta_k), \qquad T_k = \prod_{j<k} (1 - \alpha_j), \qquad w_k = T_k \, \alpha_k.$$

$$\boldsymbol{C} \approx \sum_{k=1}^{K} w_k \, \boldsymbol{c}_k \; + \; T_{K+1} \, \boldsymbol{C}_{\text{bg}}$$

Gaussian Splatting





Rendering

Each Gaussian has a global opacity parameter α_i. At a pixel p, its effective opacity becomes

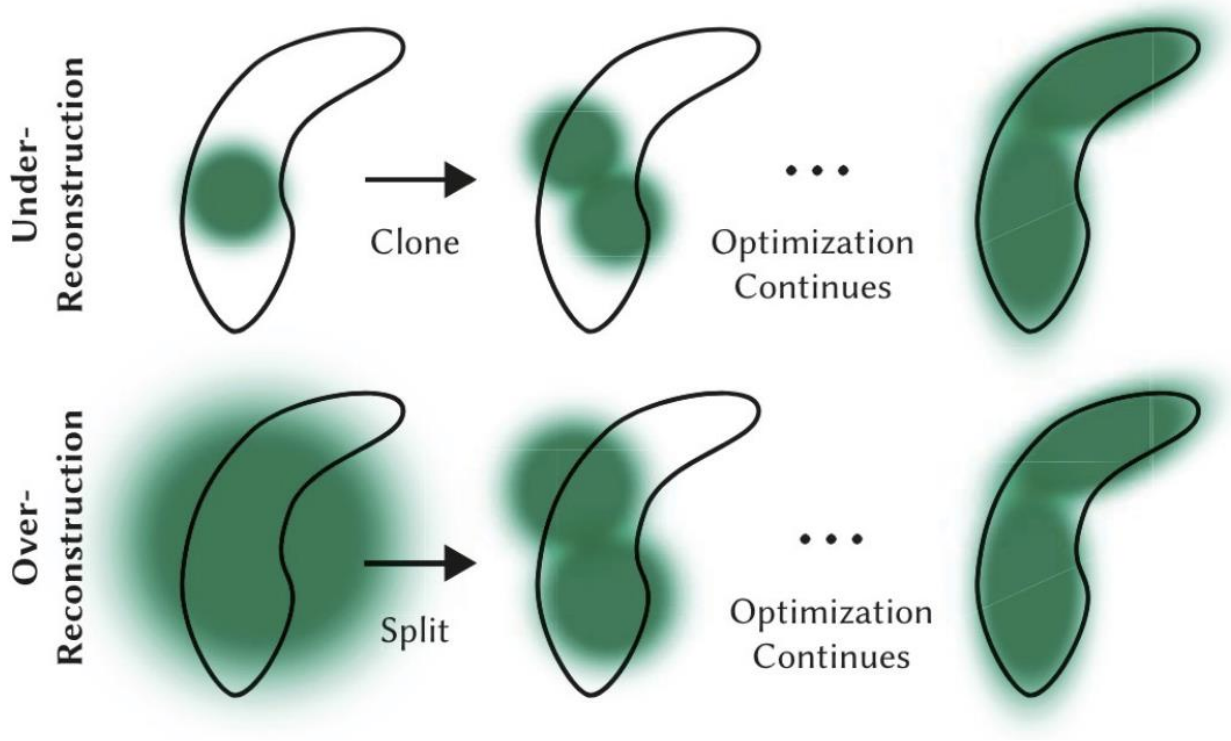
$$\boxed{\hat{\alpha}_i(p) = \alpha_i \, w_i(p)}.$$

Sort the Gaussians by depth at each pixel, because transmittance depends on order

Compute color for each pixel:

$$C = \sum_{i=1}^{N} \boldsymbol{c}_i \, \hat{\alpha}_i \prod_{j=1}^{i-1} \left(1 - \hat{\alpha}_j\right)$$

Prune and densify



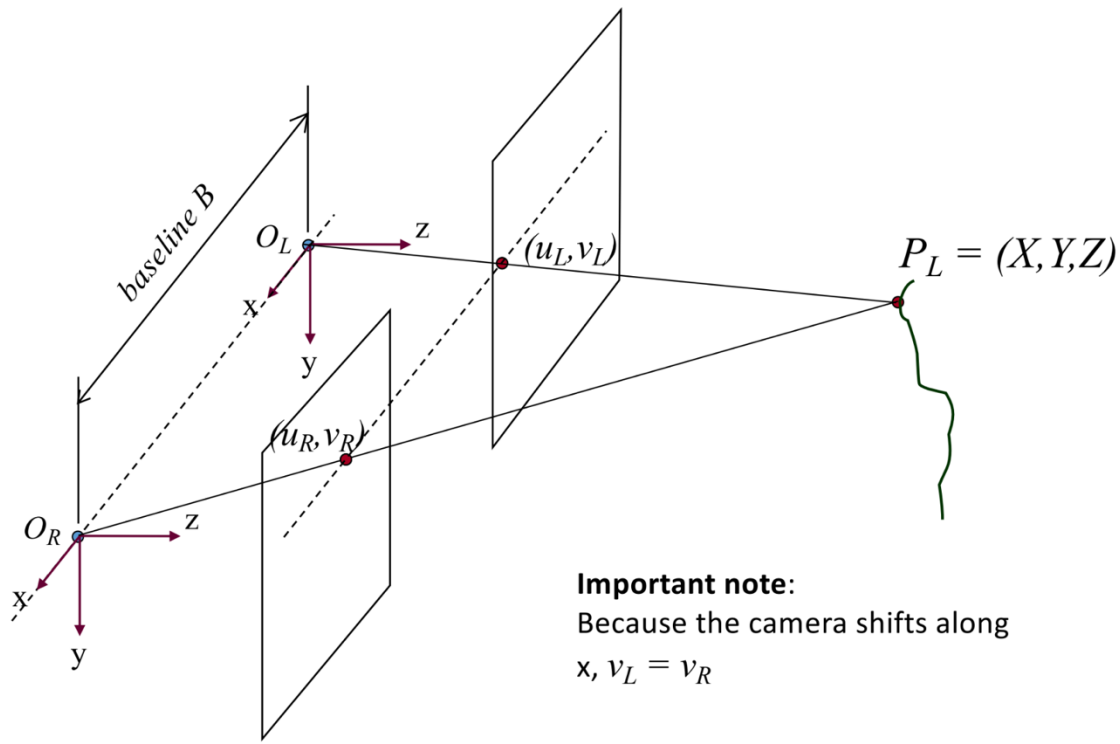
Training Loop

Repeat until convergence:

1. pick one or more training views
2. render predicted images via splatting and compositing
3. compute loss (photometric + regularizers)
4. backpropagate gradients through the renderer
5. update parameters (e.g., Adam)
6. **(every M iterations)** densify and prune Gaussians (split high-error/large-footprint splats; remove negligible splats)

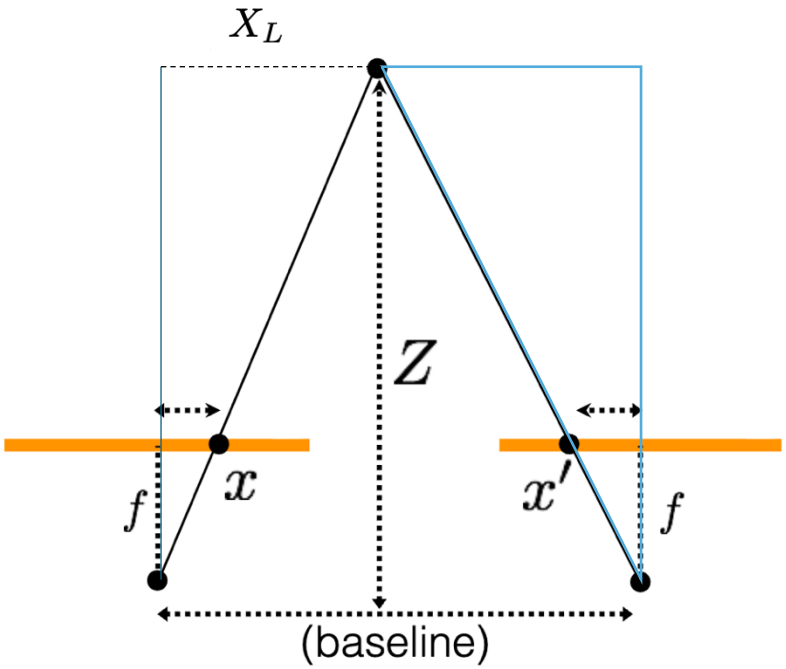
Stereo Depth Estimation II

Stereo setup: parallel image planes



Important note:
Because the camera shifts along x, $v_L = v_R$

Depth from disparity



$$\underbrace{d}_{\text{disparity}} \triangleq x - x' = \frac{fX_L}{Z} - \frac{fX_R}{Z} = \frac{f(X_L - X_R)}{Z} = \frac{fB}{Z}.$$

Rearranging yields the classic depth-from-disparity relation:

$$\boxed{Z = \frac{fb}{d}}$$

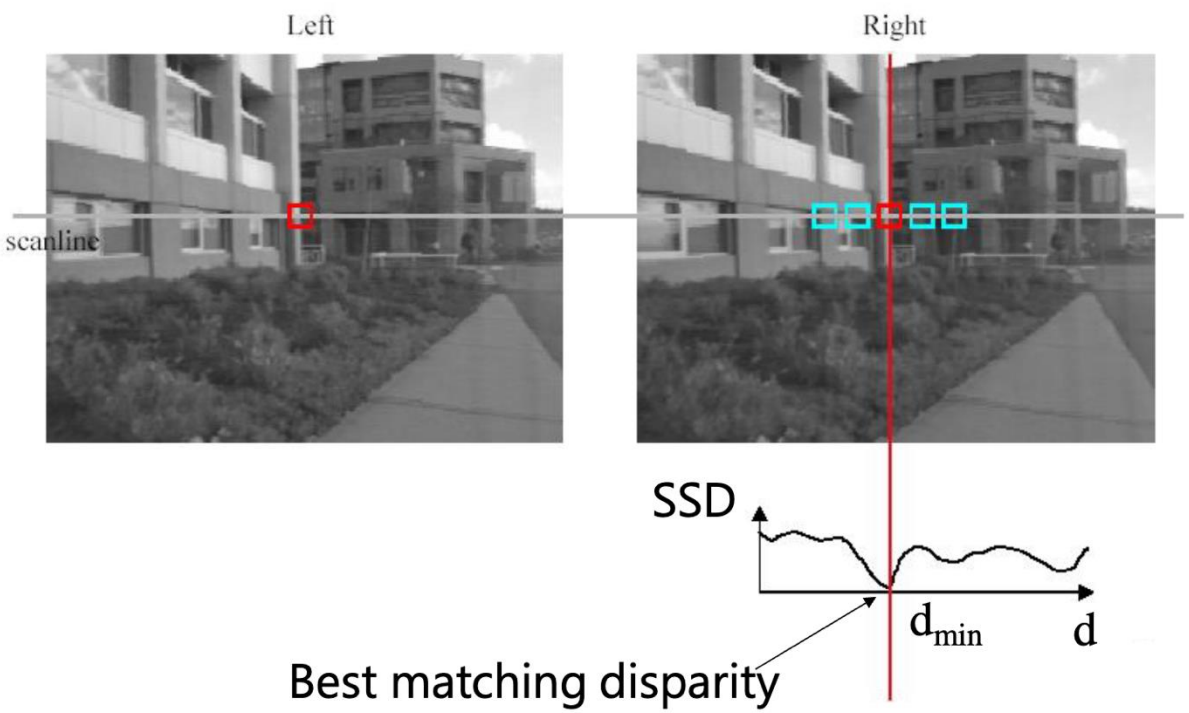
Disparity

In rectified stereo, define disparity d as the horizontal shift:

$$d(u, v) \triangleq u_L - u_R.$$

- Large disparity $\Rightarrow$ close objects
- Small disparity $\Rightarrow$ far objects
- Disparity is typically measured in pixels (possibly subpixel)

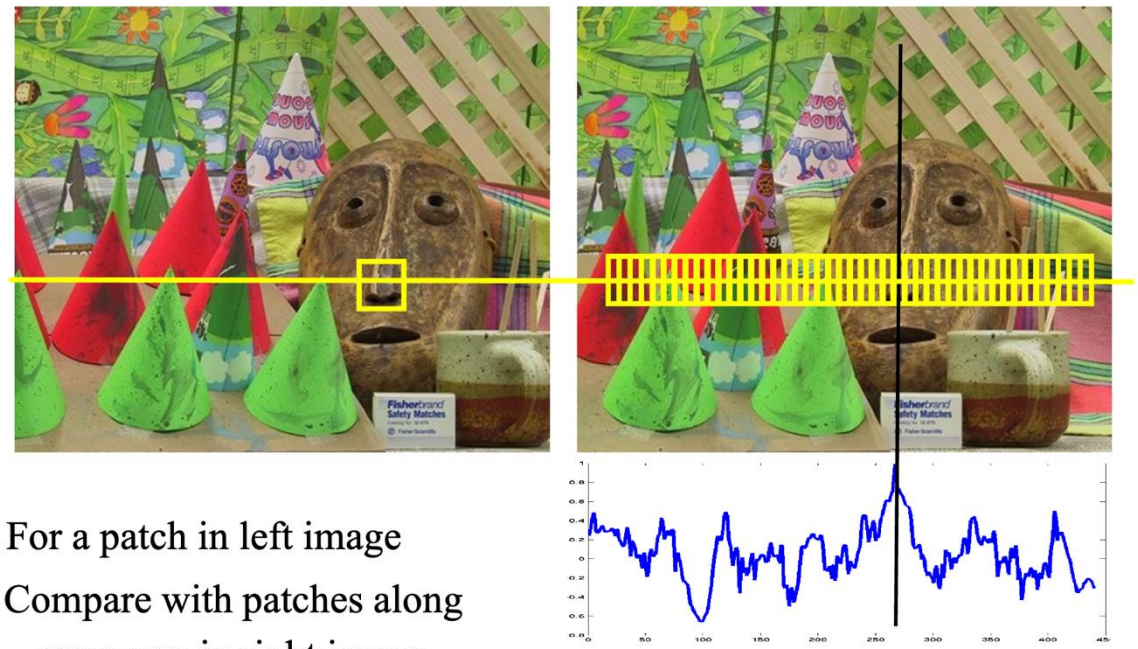
Stereo Block Matching



Matching using Epipolar Lines

Left Image

Right Image

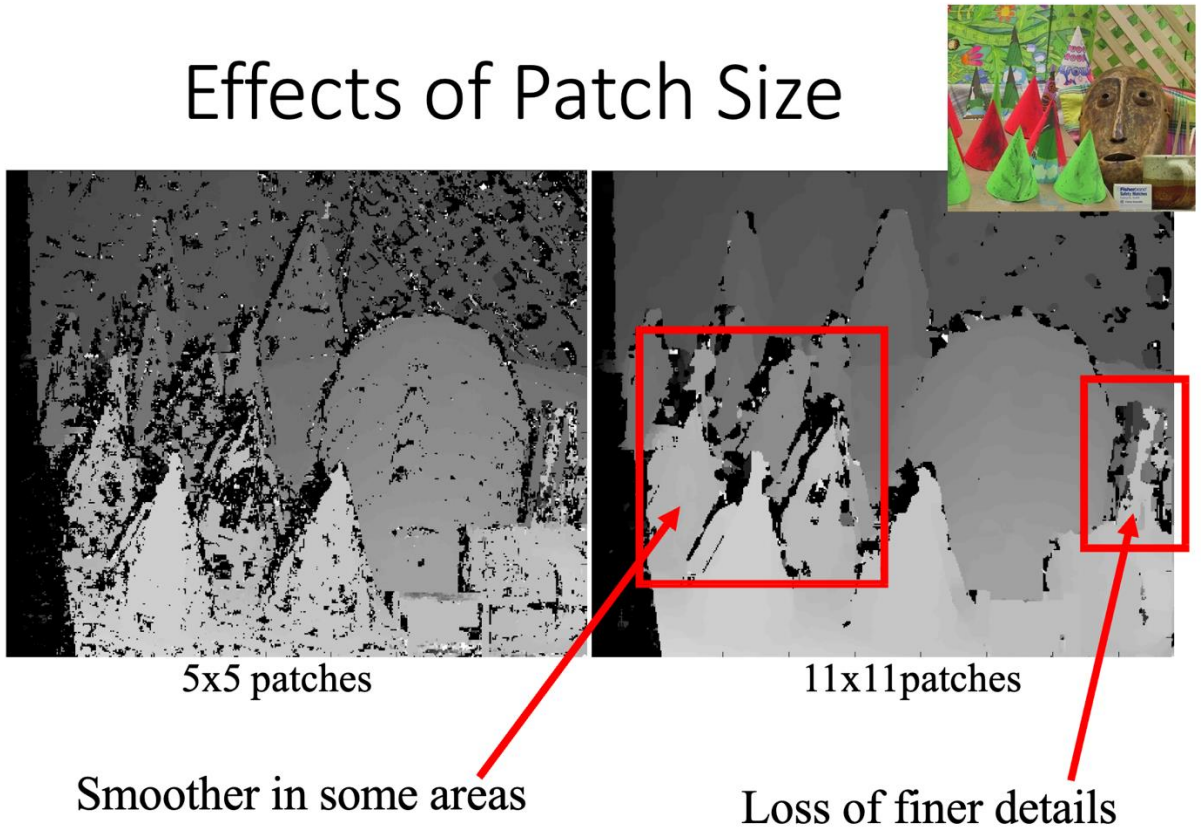


For a patch in left image

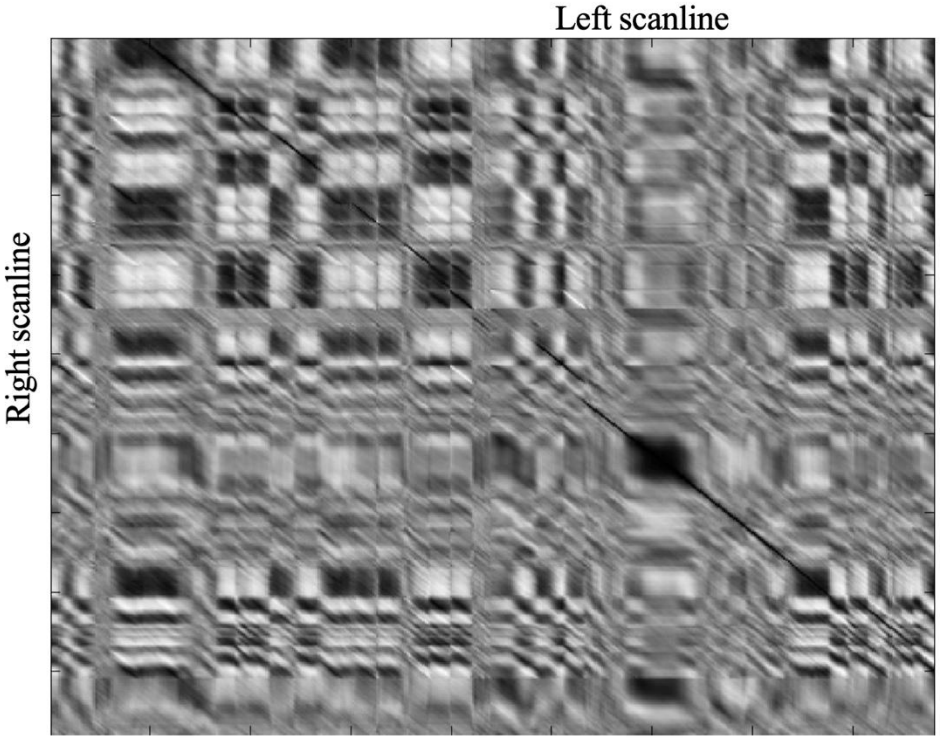
Compare with patches along same row in right image

Match Score Values (Similarity Measures)

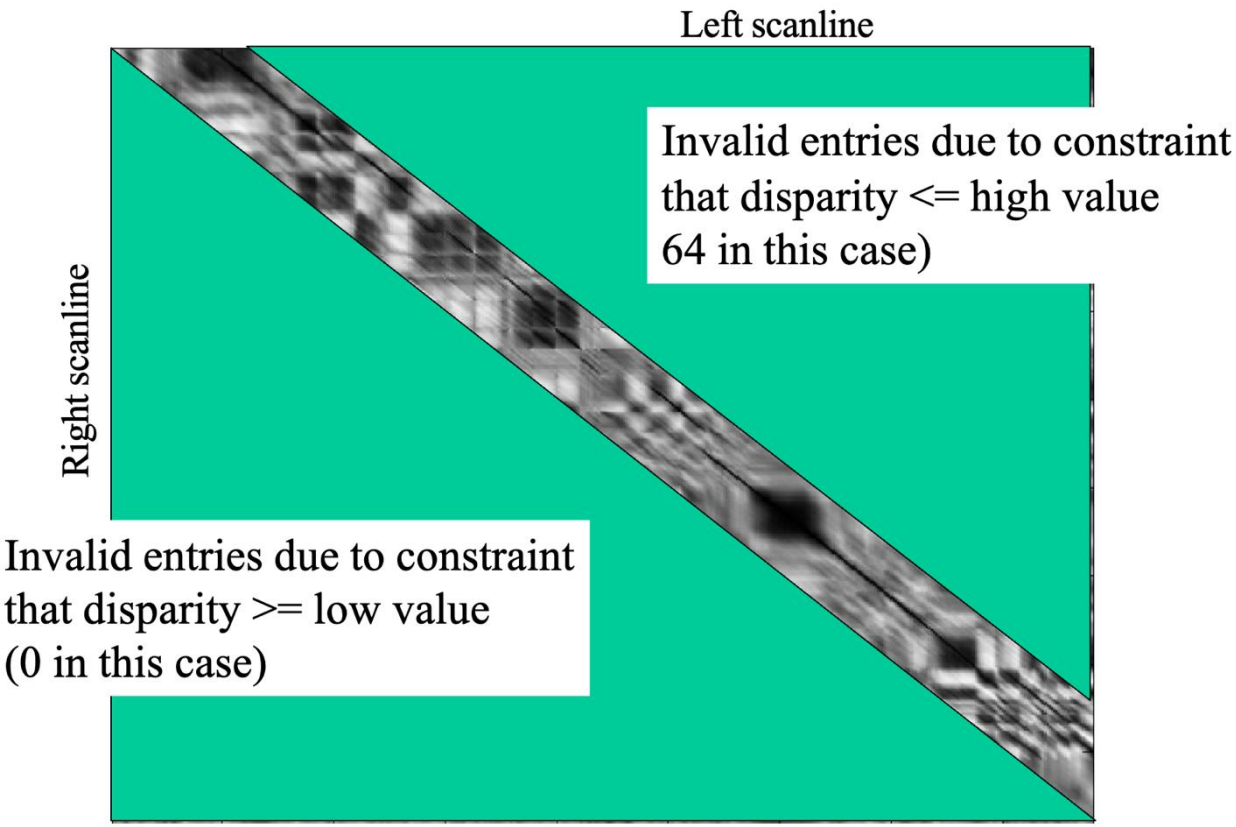
Patch Size


Effects of Patch Size
5x5 patches
11x11patches
Smoother in some areas
Loss of finer details

Disparity Space Image (DSI)

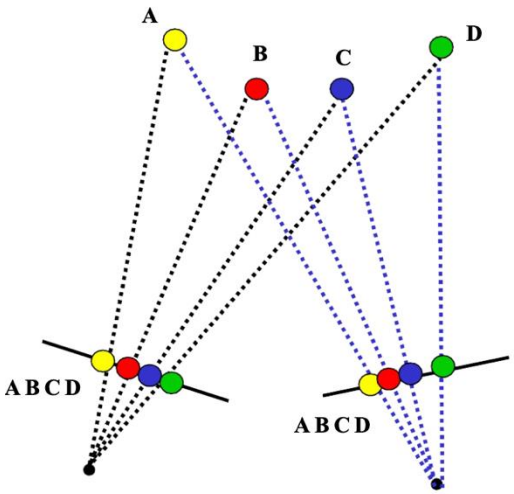


Limit range of disparity

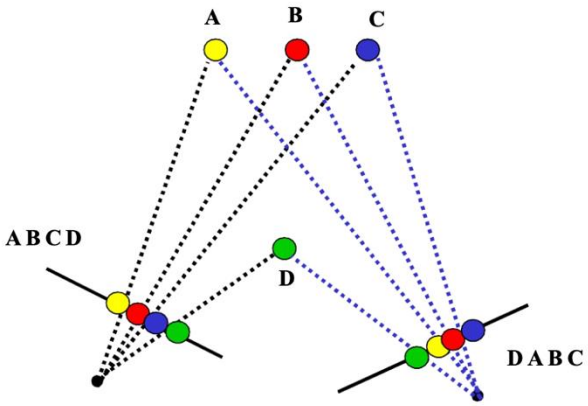


Dynamic Programming Stereo

- Ordering constraints

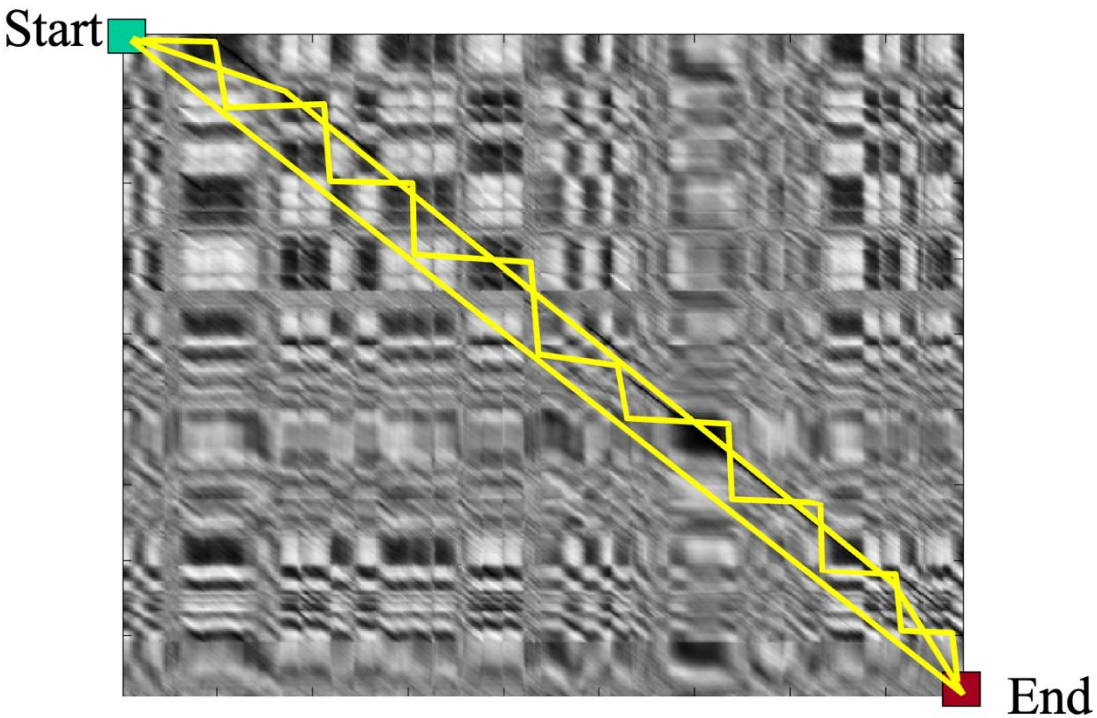


Ordering constraint…



…and its failure

DP: find a path



Finding a globally optimal monotone path

Problem Formulation

Let $i \in \{1, \ldots, W\}$ index columns in the left image (fixed row v) and $j \in \{1, \ldots, W\}$ index columns in the right image. Matching $(i \leftrightarrow j)$ has cost

$$M(i,j) \triangleq \text{cost}\big(I_L(i,v),\, I_R(j,v)\big) = C\big(i, v, \underbrace{i-j}_{d}\big).$$

We seek a **monotone path** on the (i,j) grid (often called a *disparity image*):

- **Match step**: $(i-1, j-1) \to (i,j)$ (pair pixels)
- **Occlusion step**: $(i-1, j) \to (i,j)$ or $(i, j-1) \to (i,j)$ (skip one side)

DP Stereo

Let DP(i, j) be the minimum cost to align prefixes $I_L(1{:}i, v)$ and $I_R(1{:}j, v)$.

- **Match:** pair $I_L(i, v)$ with $I_R(j, v)$

$$(i-1, j-1) \to (i, j) \quad \Rightarrow \quad \text{cost } M(i, j).$$

- **Skip right pixel (left occluded):** do not match $I_L(i, v)$

$$(i-1, j) \to (i, j) \quad \Rightarrow \quad \text{cost } \gamma.$$

- **Skip left pixel (right occluded):** do not match $I_R(j, v)$

$$(i, j-1) \to (i, j) \quad \Rightarrow \quad \text{cost } \gamma.$$

Recurrence

$$\boxed{\text{DP}(i, j) = \min\Big\{\text{DP}(i-1, j-1) + M(i, j),\ \text{DP}(i-1, j) + \gamma,\ \text{DP}(i, j-1) + \gamma\Big\}}$$

Initialize

We need base cases for the empty-prefix alignments:

$$\mathrm{DP}(0,0) = 0, \qquad \mathrm{DP}(i,0) = i\,\gamma, \qquad \mathrm{DP}(0,j) = j\,\gamma.$$

Interpretation:

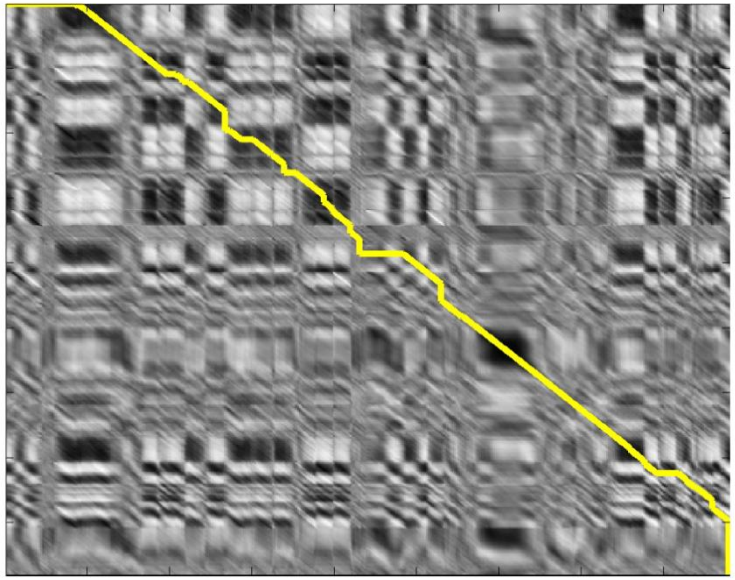
- $\mathrm{DP}(i,0)$: align i left pixels with nothing $\Rightarrow$ all are skipped (occluded)
- $\mathrm{DP}(0,j)$: align j right pixels with nothing $\Rightarrow$ all are skipped (occluded)

If you enforce a disparity range $d = i - j \in [0, D-1]$, restrict computation to the band and set

$$\mathrm{DP}(i,j) = +\infty \quad \text{for states outside the band.}$$

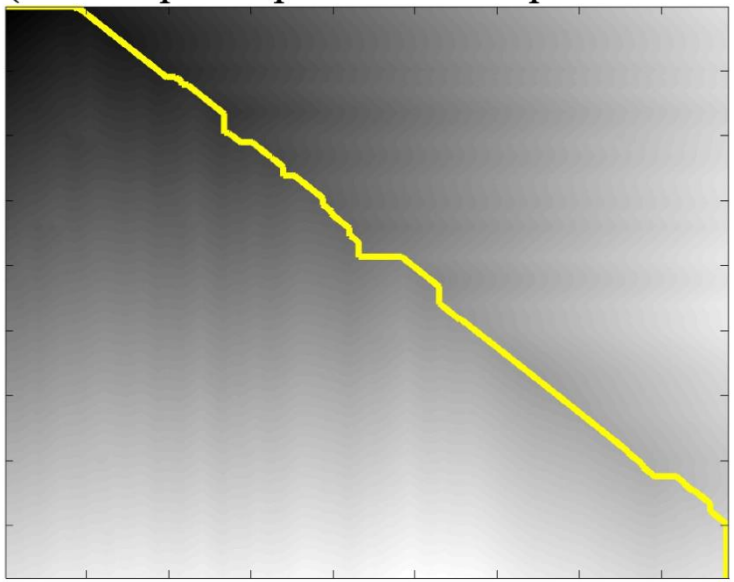
Real Example

DSI



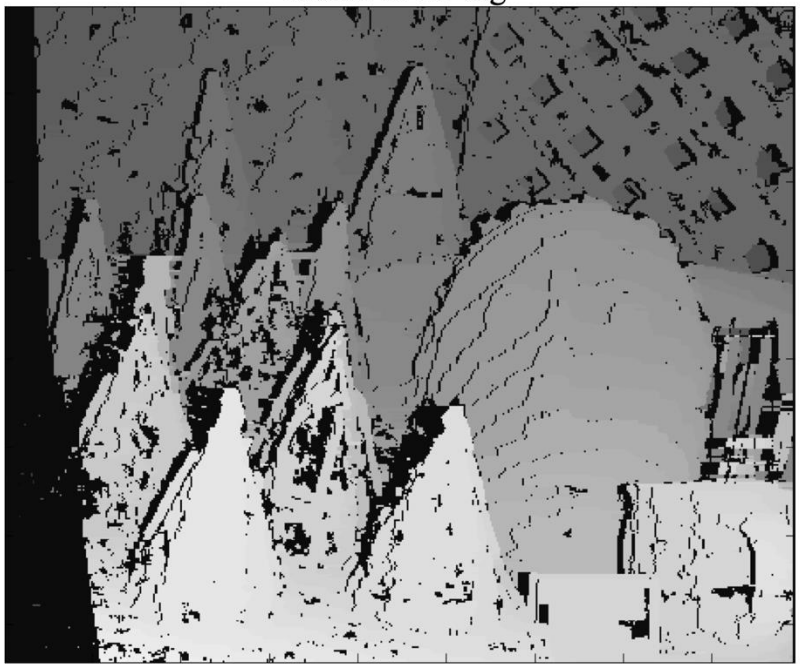
DP cost matrix

(cost of optimal path from each point to END)

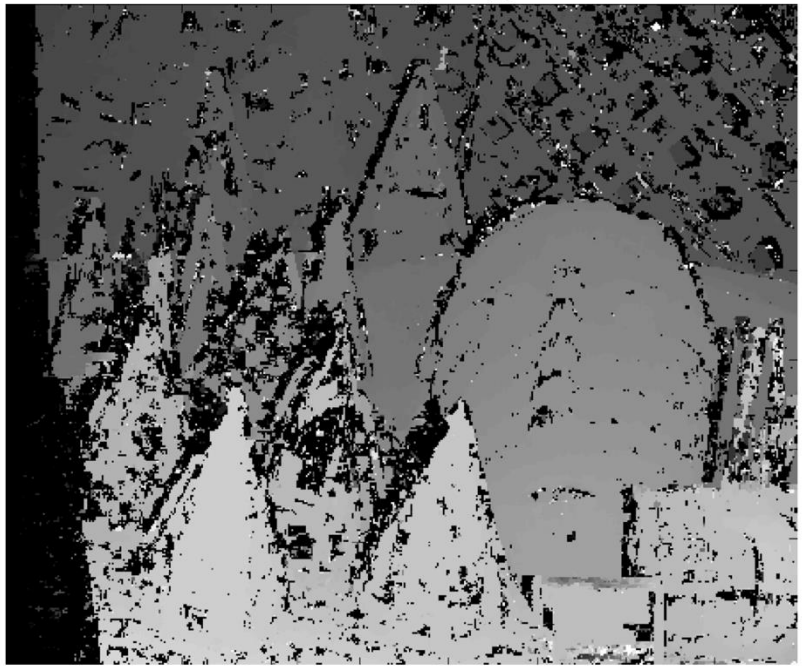


Example Results

Result of DP alg



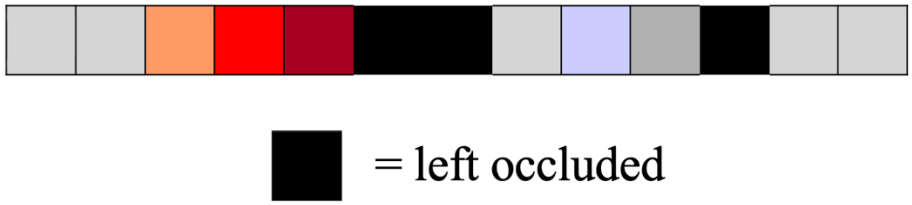
Result without DP (greedy)



Result of DP alg. Black pixels = occluded.

Occlusion Filling

Simple trick for filling in gaps caused by occlusion.

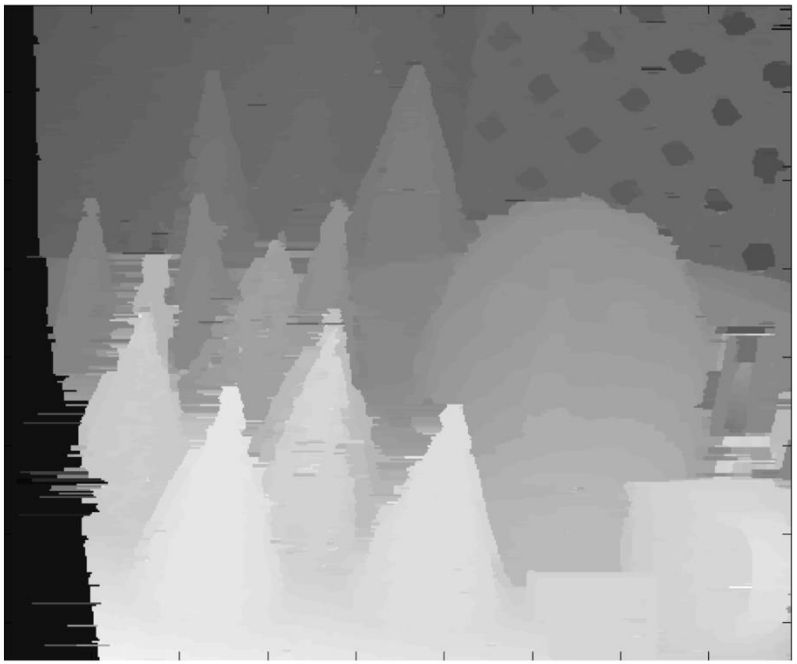


Fill in left occluded pixels with value from the nearest valid pixel preceding it in the scanline.

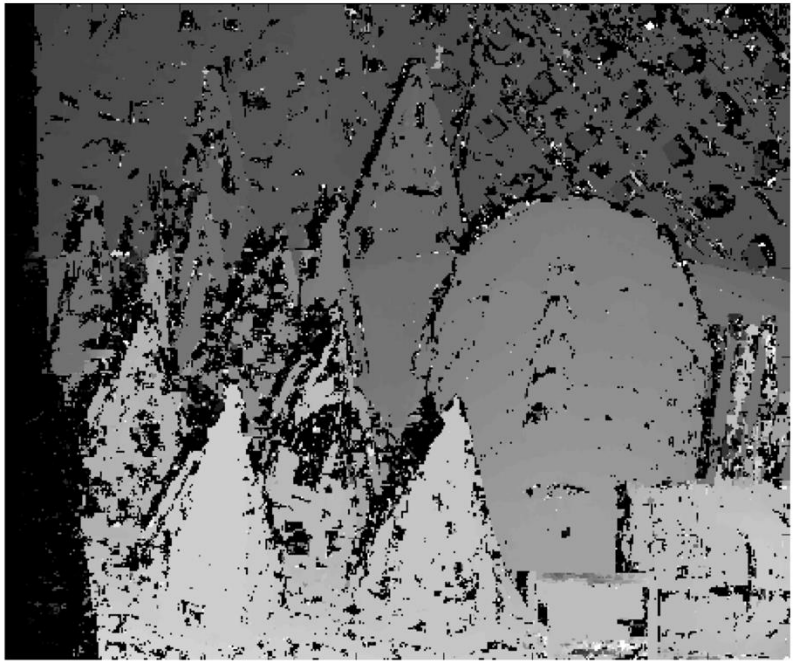
Similarly, for right occluded, look for valid pixel to the right.

Example Depth from Stereo

Result of DP alg with occlusion filling.



Result without DP (independent pixels)



Stereo as energy minimization

Data term encourages matching

$$E(d) = \boxed{\sum_{p} C(p, d(p))} + \boxed{\lambda \sum_{(p,q)\in\mathcal{N}} \rho\big(d(p) - d(q)\big).}$$

Regularization term encourages smoothness

Assumption: Depth should change smoothly

Trade-offs in designing stereo rigs

- Baseline and focal length

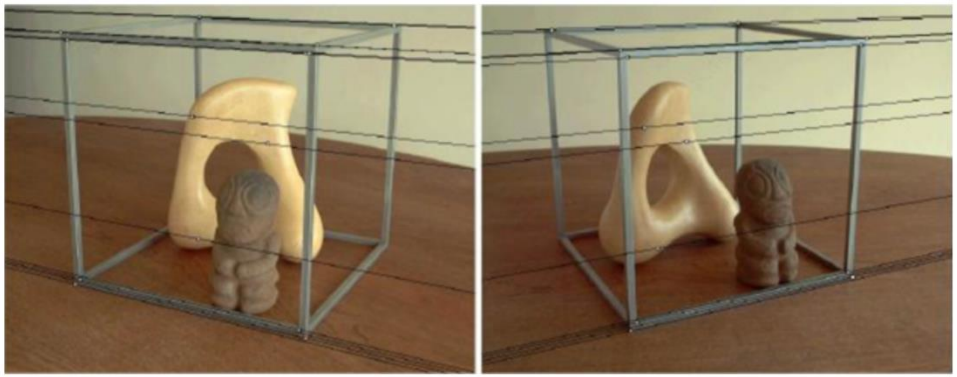
$$\frac{fb}{d^2} = \frac{fb}{(fb/Z)^2} = \frac{Z^2}{fb} \quad \Rightarrow \quad |\delta Z| \approx \frac{Z^2}{fb}|\delta d|.$$

  - Larger baseline and focal length allows same depth change to be more visible in disparity

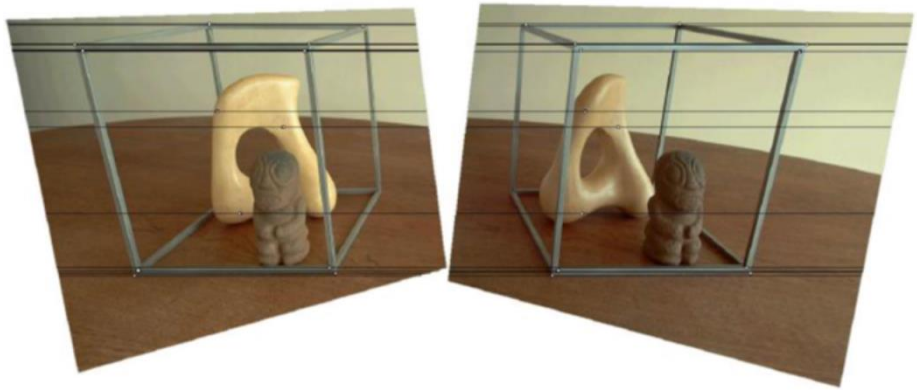
- Large baseline: more occlusion

- Longer focal length: smaller FOV

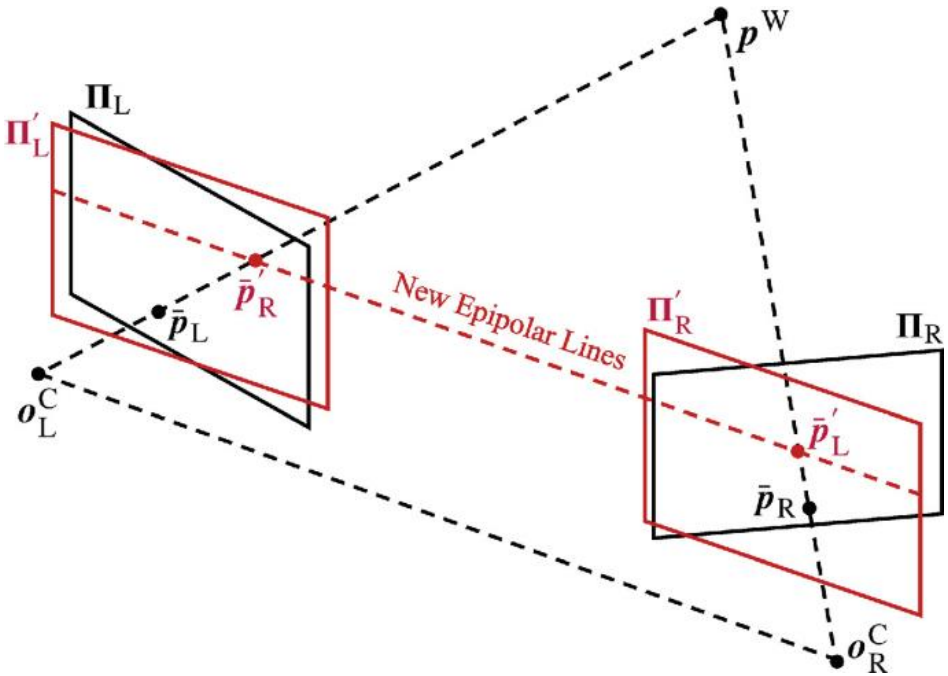
Rectification



Horizontal epipolar line

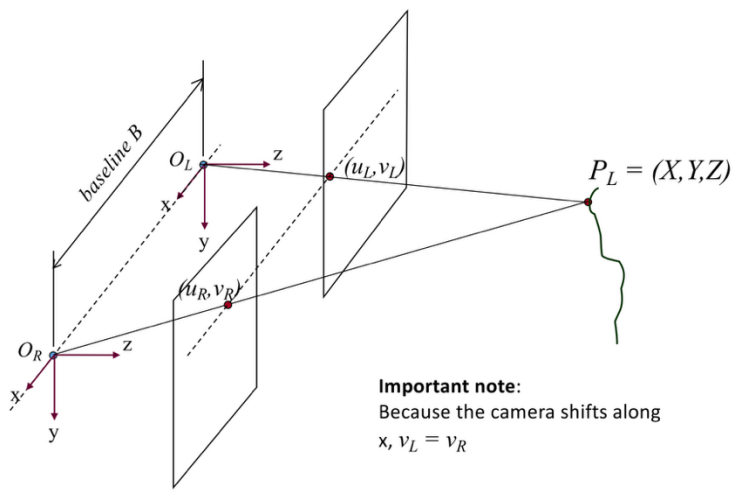


Rectification is just rotating both cameras



Requirements for rectification

- Image x direction is parallel to baseline
- Y and Z directions are shared between the two frames



Why rectification?

- Epipolar lines become horizontal

$$R_{\text{rel}} = I, \qquad t = (b, 0, 0)^\top.$$

For a point $x_L = (u_L, v_L, 1)^\top$, its epipolar line in the right image is

$$\ell_R = E\, x_L = [t]_\times x_L = t \times x_L = (0,\, -b,\, bv_L)^\top.$$

The epipolar constraint is $\ell_R^\top x_R = 0$; with $x_R = (u_R, v_R, 1)^\top$:

$$(0)u_R + (-b)v_R + (bv_L) \cdot 1 = 0 \;\Rightarrow\; v_R = v_L.$$

Therefore correspondences lie on the same scanline: **horizontal epipolar lines**.

Constructing rectified axes

- Construct x so that it is the direction of the baseline
- Find reasonable y and z axes that are
  - Orthogonal with each other and with x
  - Point forward

Compute axes of a rectified frame

1) Baseline defines the rectified x axis

$$\hat{\mathbf{x}} = \frac{\mathbf{t}_{LR}}{\|\mathbf{t}_{LR}\|}.$$

2) Average the cameras' vertical (image-down) direction

$$\boxed{\mathbf{y}_{\text{avg}} = \frac{\mathbf{e}_y + \mathbf{R}_{LR}^\top \mathbf{e}_y}{\|\mathbf{e}_y + \mathbf{R}_{LR}^\top \mathbf{e}_y\|}.}$$

Not necessarily orthogonal to x

Compute axes of a rectified frame

3) Compute the rectified optical axis z

$$\hat{\mathbf{z}} = \frac{\hat{\mathbf{x}} \times \hat{\mathbf{y}}}{\|\hat{\mathbf{x}} \times \hat{\mathbf{y}}\|}.$$

4) Recompute y to enforce exact orthogonality

Recompute

$$\hat{\mathbf{y}} \leftarrow \frac{\hat{\mathbf{z}} \times \hat{\mathbf{x}}}{\|\hat{\mathbf{z}} \times \hat{\mathbf{x}}\|}.$$

Now $(\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}})$ is a right-handed orthonormal basis (up to normalization).

Pixel mapping is through homography

Define the rectified rotation (mapping *left camera* coordinates into rectified coordinates):

$$\boxed{\mathbf{R}_{\text{rect}} = \begin{bmatrix}\hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}}\end{bmatrix}^\top}$$

Then both cameras are rotated into the same rectified axes by

$$\boxed{\mathbf{R}_L = \mathbf{R}_{\text{rect}}, \qquad \mathbf{R}_R = \mathbf{R}_{\text{rect}}\,\mathbf{R}_{LR}^\top}$$

3D rotations into 2D image warps via homographies

$$\boxed{\mathbf{H}_L = \mathbf{K}'\,\mathbf{R}_L\,\mathbf{K}_L^{-1}, \qquad \mathbf{H}_R = \mathbf{K}'\,\mathbf{R}_R\,\mathbf{K}_R^{-1}}$$

Why?

Recap: homography relates two rotated views

Use the (pinhole) projection equation for a 3D point $\boldsymbol{X}$:

$$\lambda\, \tilde{\boldsymbol{x}} = \mathbf{K}\,[\mathbf{R} \mid \mathbf{t}]\, \tilde{\boldsymbol{X}}.$$

Consider two views of the same camera center (**pure rotation**, so $\mathbf{t}_1 = \mathbf{t}_2 = \mathbf{0}$), with the same intrinsics $\mathbf{K}$:

$$\lambda_1\, \tilde{\boldsymbol{x}}_1 = \mathbf{K}\mathbf{R}_1\, \boldsymbol{X}, \qquad \lambda_2\, \tilde{\boldsymbol{x}}_2 = \mathbf{K}\mathbf{R}_2\, \boldsymbol{X}.$$

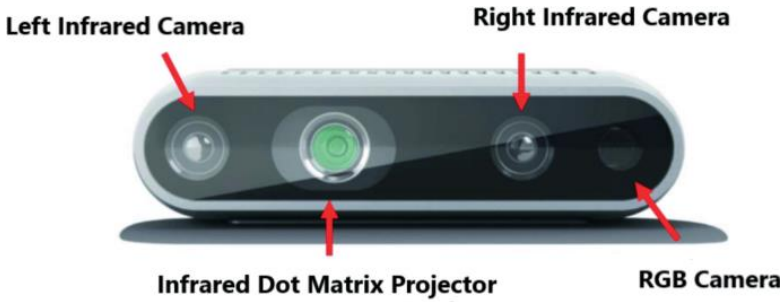
Eliminate $\boldsymbol{X}$ using the first equation:

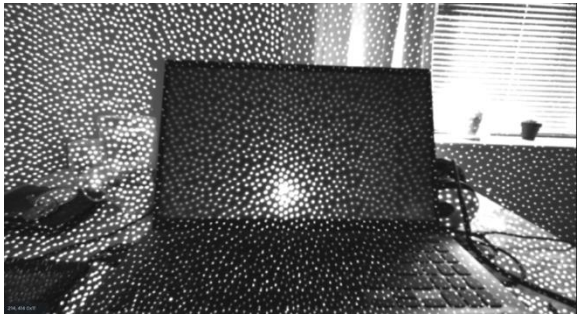
$$\boldsymbol{X} = \mathbf{R}_1^\top \mathbf{K}^{-1}(\lambda_1 \tilde{\boldsymbol{x}}_1).$$

Substitute into the second:

$$\lambda_2\, \tilde{\boldsymbol{x}}_2 = \mathbf{K}\mathbf{R}_2\mathbf{R}_1^\top \mathbf{K}^{-1}(\lambda_1 \tilde{\boldsymbol{x}}_1) \quad \Rightarrow \quad \tilde{\boldsymbol{x}}_2 \sim \underbrace{\mathbf{K}\mathbf{R}_2\mathbf{R}_1^\top \mathbf{K}^{-1}}_{\mathbf{H}}\, \tilde{\boldsymbol{x}}_1.$$

Real-time stereo





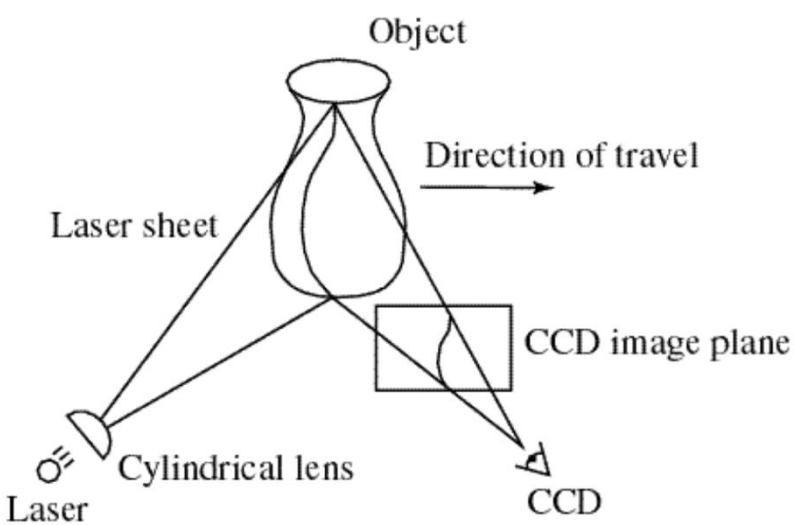
Li Zhang's one-shot stereo

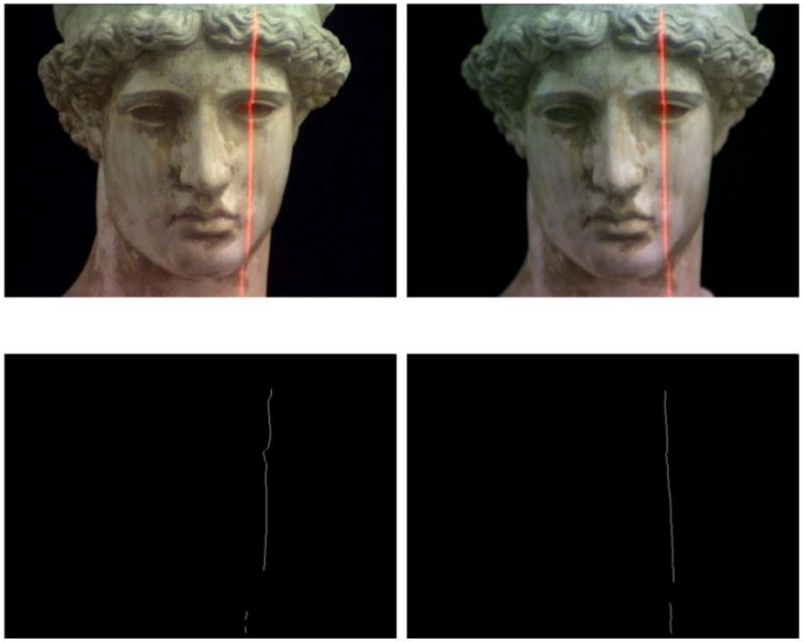
https://www.cs.cornell.edu/courses/cs5670/2022sp/lectures/lec13_stereo_for_web.pdf

Active Stereo

https://ios.gadgethacks.com/news/watch-iphone-xs-30k-ir-dots-scan-your-face-0180944/

Controlled Structured Light





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